Workout Video Recommendation System – Proposed Approach

Overview

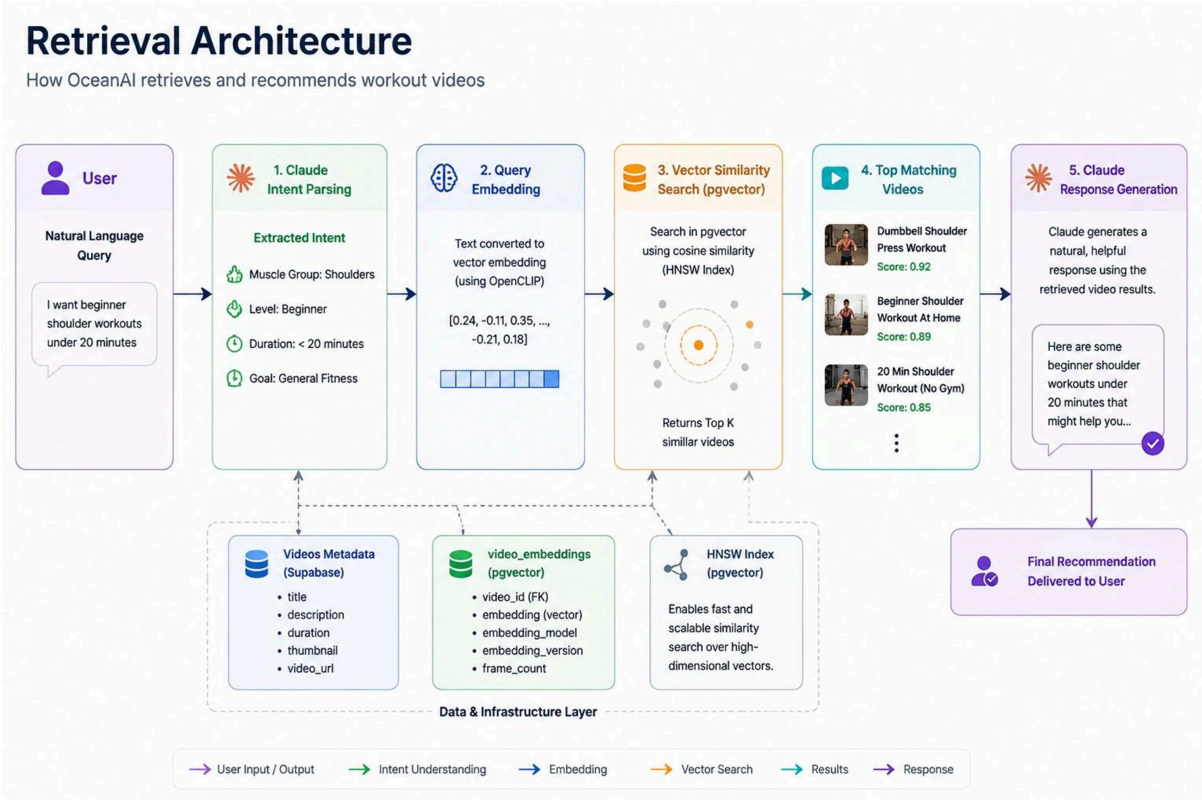
Based on our discussion, my plan is to build a semantic workout video recommendation system that does not depend entirely on titles, descriptions, or manually added tags.

The main idea is to generate embeddings directly from the visual content of each workout video so that recommendations are based on what is actually happening inside the video rather than only on metadata.

High-Level Architecture

The overall workflow is:

Video Upload
↓
Supabase Storage
↓
Videos Table
↓
Supabase Webhook
↓
Embedding Processor
↓
OpenCLIP
↓
Hybrid Embedding
↓
video_embeddings
↓
Vector Retrieval
↓
Claude
↓
User Recommendation



Architecture Diagram

This architecture ensures that every newly uploaded video automatically becomes searchable without requiring any manual processing.

Database Design

Instead of modifying the existing videos table, I am proposing a separate video_embeddings table.

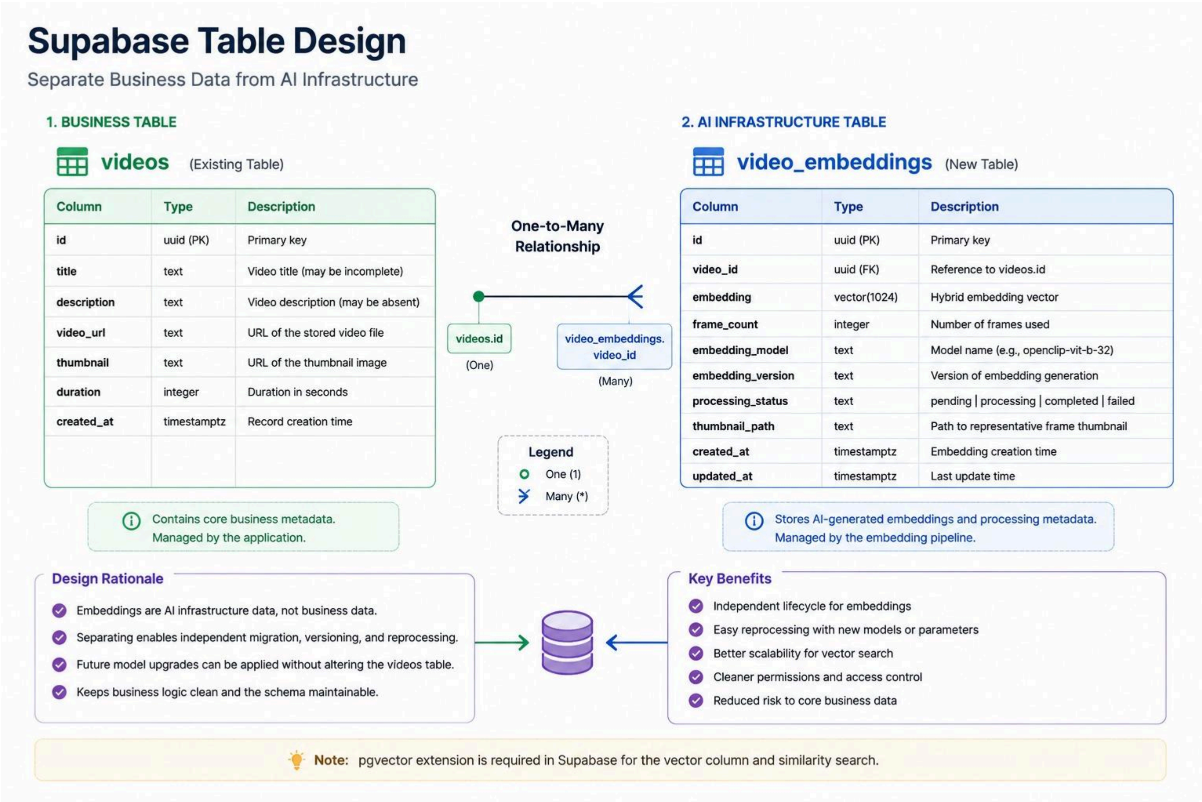
The existing videos table will continue storing business-related information such as:

- Title
- Description
- Video URL
- Thumbnail
- Duration

The new video_embeddings table will store AI-specific information such as:

- Video ID
- Embedding Vector
- Frame Count
- Embedding Model

- Embedding Version
- Processing Status
- Created/Updated Timestamp



Supabase table design

This separation keeps business data and AI infrastructure independent, making future upgrades, reprocessing, and migrations much easier.

Embedding Generation Strategy

For video understanding, I plan to use OpenCLIP.

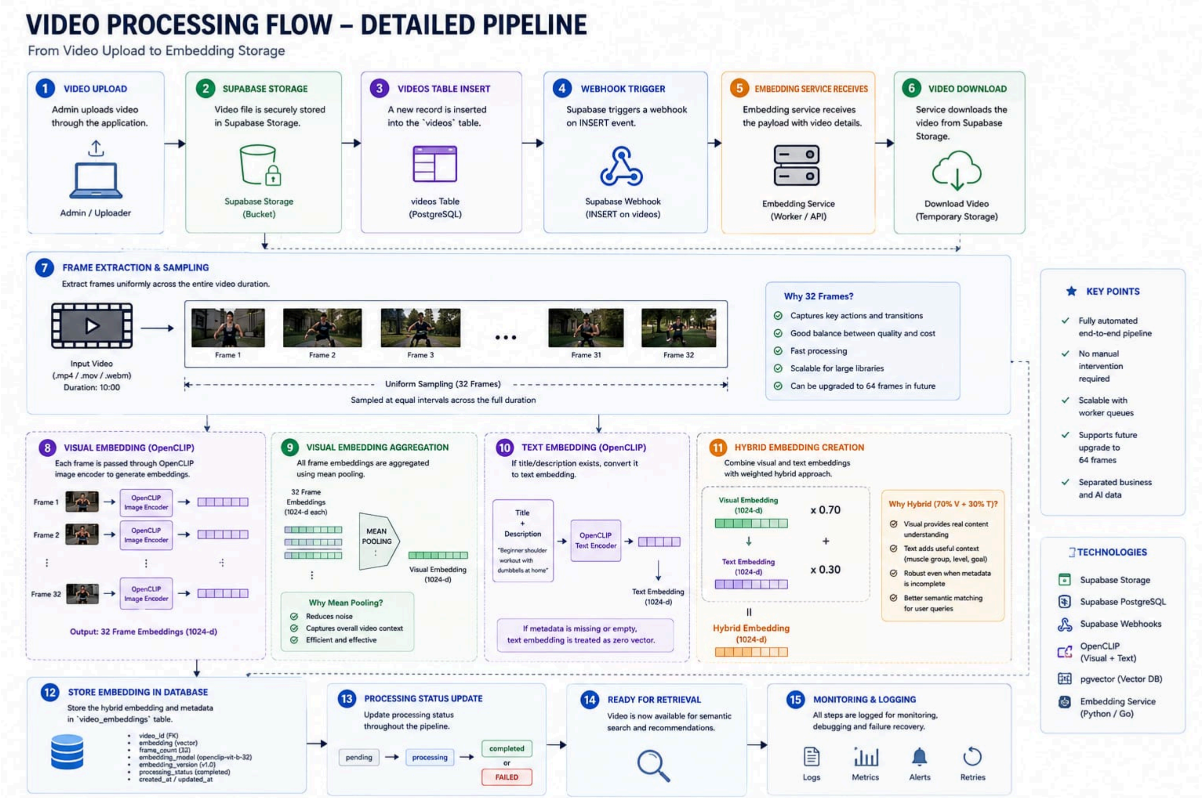
When a video is uploaded, frames will be extracted uniformly across the video's duration. Currently, I am planning to use **32 frames per video** because it provides a good balance between quality, processing cost, and retrieval performance.

Although 64 frames can provide slightly better visual understanding, 32 frames is a safer starting point and can be increased later if required.

The processing flow will be:

Video
↓
Frame Extraction (32 Frames)
↓
OpenCLIP Visual Encoder
↓
Frame Embeddings
↓
Aggregation
↓
Final Visual Embedding

Each extracted frame will generate an embedding, and all frame embeddings will be combined into a single representation of the video.



Video processing flow

Hybrid Embedding Approach

I do not want recommendation quality to depend entirely on titles and descriptions because metadata can often be incomplete or inaccurate.

If metadata is available, it will be used as an additional signal.

The final embedding will be generated using:

- 70% Visual Embedding
- 30% Text Embedding

This means the recommendation system primarily understands the actual workout content while still benefiting from useful metadata when available.

The goal is to make recommendations accurate even when titles, tags, or descriptions are missing.

Automated Processing

The embedding pipeline will be fully automated using Supabase Webhooks.

Whenever a new record is inserted into the videos table:

1. Supabase triggers a webhook.
2. The embedding service receives the event.
3. The uploaded video is downloaded.
4. Frames are extracted.
5. OpenCLIP generates embeddings.
6. The final embedding is stored in video_embeddings.
7. The video becomes available for retrieval.

No manual trigger or admin action will be required.

Retrieval Strategy

When a user submits a query, the same OpenCLIP model will convert the query into an embedding.

Examples:

- Beginner shoulder workouts
- Fat loss exercises
- Mobility routines
- Chest workouts without equipment

The query embedding will then be compared against all stored video embeddings using pgvector similarity search.

This allows retrieval based on semantic meaning rather than exact keyword matching.

Claude Integration

Claude will not be responsible for retrieving videos.

Its responsibility will be limited to:

Intent Understanding

Example:

User Query:

> I want beginner shoulder workouts

Extracted Intent:

```
{
  "muscle_group": "shoulders",
  "level": "beginner"
}
```

Response Generation

After retrieval, Claude will receive the matching videos and generate a natural response explaining the recommendations.

This keeps retrieval deterministic while allowing responses to remain conversational and user-friendly.

Recommendation Flow

User Query
↓
Intent Understanding
↓
Query Embedding
↓
Vector Similarity Search
↓
Top Matching Videos
↓
Claude Response Generation
↓
Final Recommendation

When multiple videos match a query, ranking will be performed using semantic similarity scores rather than simple metadata filtering.

This should significantly improve recommendation quality as the video library grows.

Expected Outcome

The final system will ensure that:

- Every uploaded video automatically becomes searchable.
- Recommendations are based primarily on actual video content.
- Metadata acts as a supporting signal rather than the primary source of truth.
- Recommendation quality improves as the video library grows.
- Future model upgrades can be introduced without major database changes.

Overall, this approach provides a scalable foundation for AI-powered workout video recommendations while keeping the architecture clean, maintainable, and easy to extend in the future.